NEOSTATS · AI ENGINEER ROUND 1

AI-Powered Credit Risk Intelligence Platform

EDA · ML risk scoring · explainability · talk-to-data · business rules · deployed end-to-end

Built on the Kaggle Home Credit Default Risk dataset (307,511 applications)

Meet Virani

THE BRIEF

What this platform had to do

- Score loan applicants for default risk using a trained ML model, with explicit handling of class imbalance (only **8.07%** of applicants default)
- Explain individual predictions in clear, human-readable terms
- Let a non-technical user ask questions of the data in plain English
- Turn the ML model into business-readable IF-THEN rules a policy team can use
- Deliver the platform as a deployed, usable web application

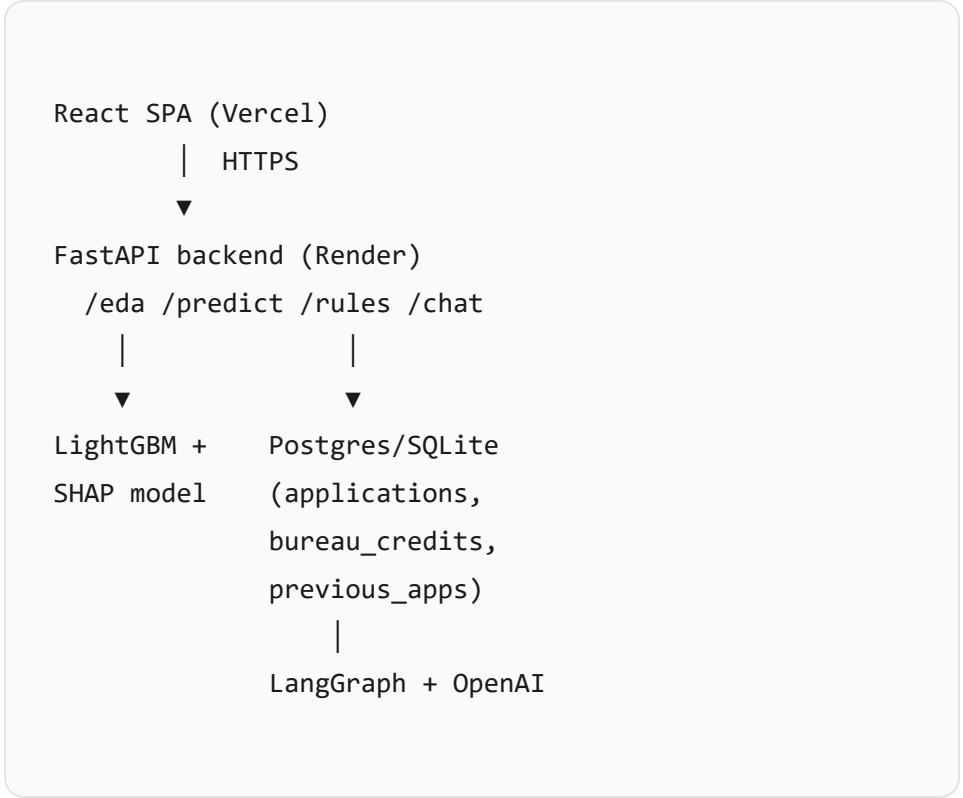
Approach in one line

One LightGBM model is trained once and reused throughout the platform: for scoring, for SHAP-based explanations, and as the basis for a surrogate decision tree that produces the business rules. A single database powers a LangGraph SQL agent so that every chatbot answer is grounded in real, queryable data.

SYSTEM DESIGN

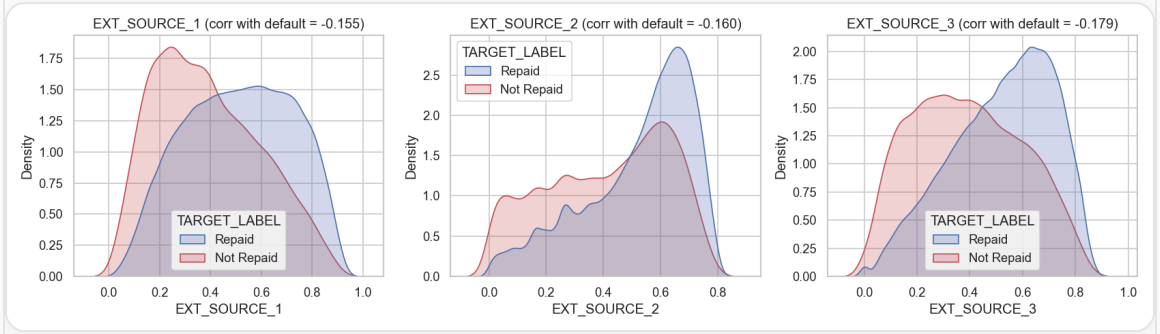
Architecture

- **React SPA (Vercel)**, with 5 pages: Data Overview, Risk Prediction, Explainability, Business Rules, and Ask the Data
- **FastAPI backend (Render)**, a thin HTTP layer over the ML and data modules, exposing `/eda`, `/predict`, `/rules`, and `/chat`
- **LightGBM model with SHAP**, trained offline and loaded as `model.joblib` to score applicants and explain each prediction
- **Postgres (Neon) or SQLite**, using the same schema behind a single `DATABASE_URL` setting, powering the talk-to-data chatbot
- **LangGraph with OpenAI**: rewrite, generate SQL, validate, execute, and summarize



5 findings that shaped every downstream decision

- **EXT_SOURCE_1/2/3** are the strongest single predictors, 20-30% missing, so LightGBM's native missing-value handling mattered
- **Age**: 20-30 year-olds default at roughly 2.3x the rate of 60-70 year-olds
- **Income type** is a strong differentiator, especially at the extremes (Working ~9.6% vs Pensioner ~5.4%)
- **Debt burden** ratios (credit/income, annuity/income) separate the two classes, motivating engineered features
- **Bureau history**: applicants with no prior credit bureau record default more (10.1% vs 7.7%), justifying the bureau.csv join



LightGBM, with scale_pos_weight for imbalance

Why LightGBM

- Handles missing values natively, no lossy imputation needed for EXT_SOURCE_*
- Handles categoricals directly, no one-hot blowup
- Trains on 307k rows in under a minute on a laptop CPU

Handling the 8.07% default rate

- `scale_pos_weight` ≈ 11.4 (negative:positive ratio), reweights the loss instead of resampling
- Evaluated using ROC-AUC and PR-AUC rather than accuracy, since a model that always predicts "repaid" would score 92% accuracy while offering no real predictive value

Note on model training

During training, early stopping was tracking LightGBM's default `binary_logloss` metric instead of the intended `auc` metric. Training completed without errors but stopped after only 2 boosting rounds, while the true validation AUC was still climbing, from 0.72 to 0.76 by round 102. This was identified by inspecting `evals_result_` and resolved by setting `metric="auc"` on the estimator itself along with `first_metric_only=True`.

Final: 661 boosting rounds, ROC-AUC 0.7725

SECTION 4 · EVALUATION METRICS & RESULTS

Validated on a held-out 61,502-row split

0.7725
ROC-AUC

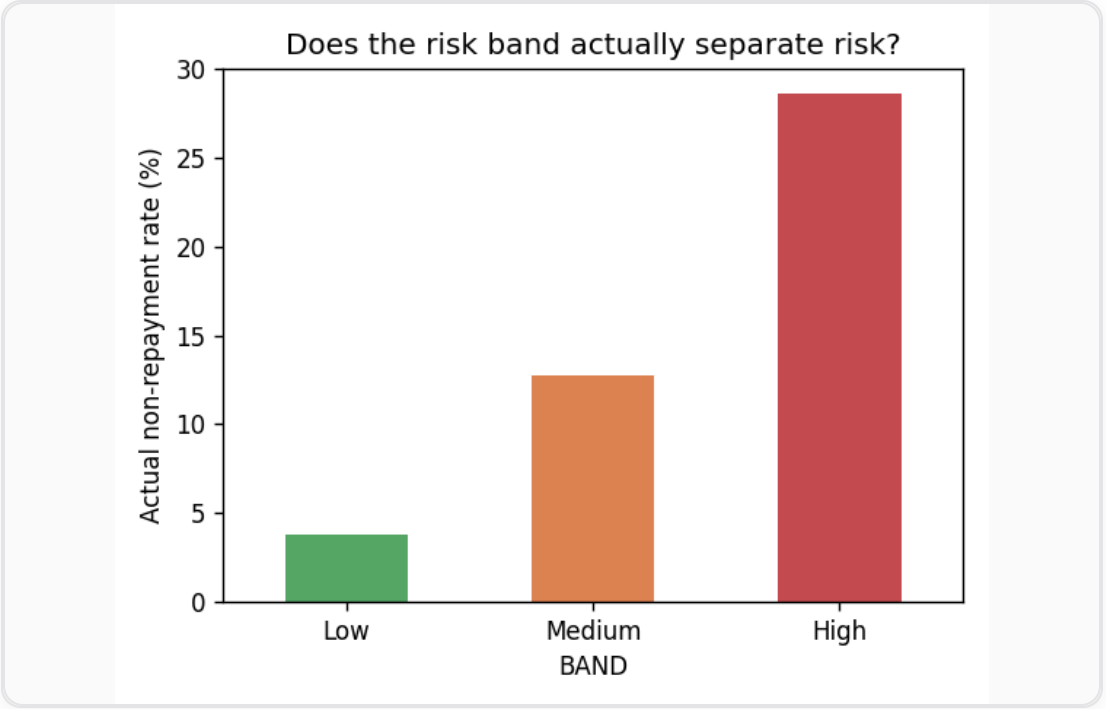
0.2629
PR-AUC (vs 0.081 random baseline)

661
Best boosting round

Risk bands, validated against real outcomes

BAND	APPLICANTS	ACTUAL NON-REPAYMENT RATE
Low	43,051	**3.8%**
Medium	12,300	**12.8%**
High	6,151	**28.6%**

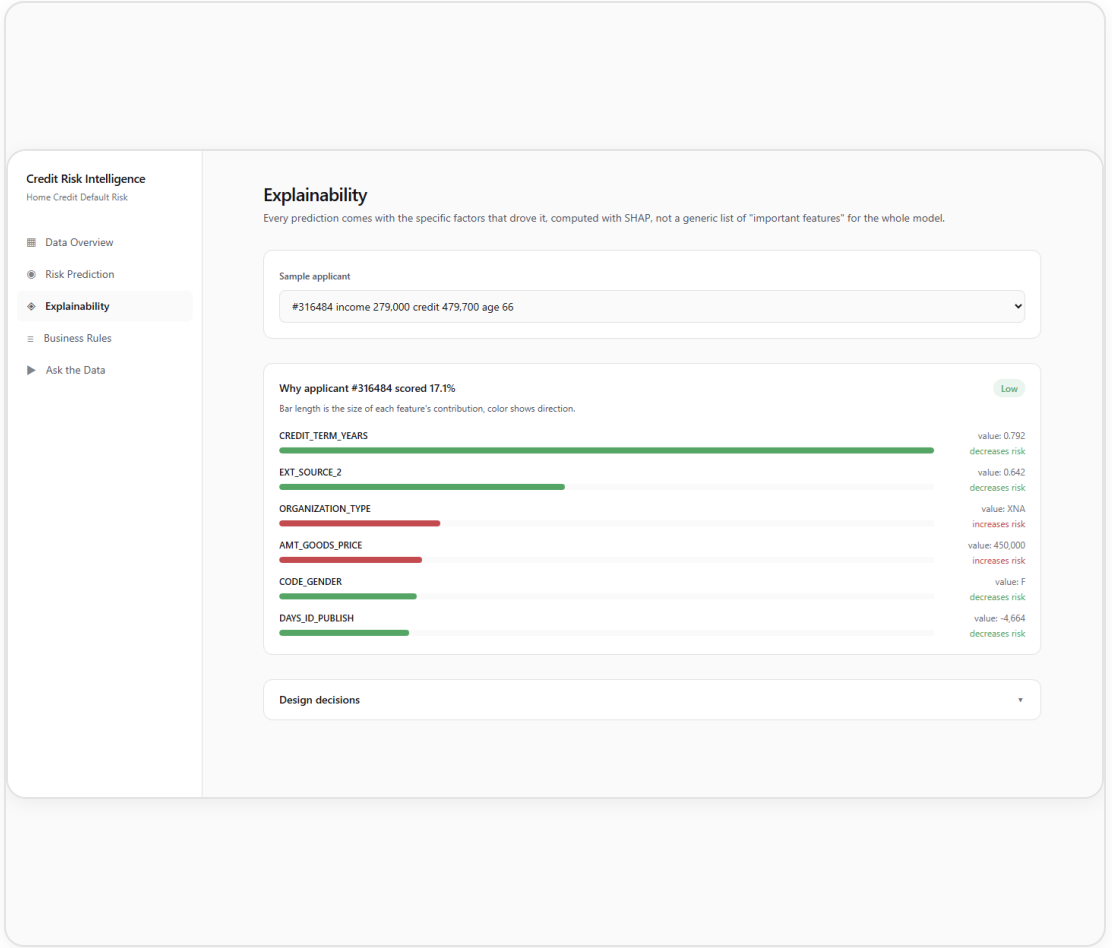
A 7.5x gap between the Low and High bands shows that they separate risk meaningfully. The bands are percentile-based (top 10% / next 20% / rest) rather than fixed probability cutoffs, since fixed cutoffs would leave "High" nearly empty at an 8% base rate.



SECTION 3-4 · EXPLAINABILITY

Per-applicant explanations using SHAP

- Every prediction ships with a **TreeExplainer** breakdown of the specific factors that drove that specific score
- Each factor shows its direction (increases / decreases risk) and its actual value for that applicant
- This is the model-level counterpart to the surrogate rules: SHAP explains one prediction, rules summarize the model's overall policy
- Top global drivers: `ORGANIZATION_TYPE`, `CREDIT_TERM_YEARS` (engineered), `EXT_SOURCE_3/1/2`, `OCCUPATION_TYPE`, and a bureau-derived feature ranking 7th overall



SECTION 6 · BUSINESS RULES

Distilling the model into 3 business rules

A shallow surrogate decision tree approximates LightGBM's own scores, and every root-to-leaf path is then validated against **real held-out outcomes** rather than the surrogate's own approximation. Rules that do not hold up under this validation are dropped, which removed 3 of the 6 candidate rules.

Design tradeoff: the EXT_SOURCE_* scores, the model's strongest predictors, are excluded from the surrogate by design, since an external score is not something a policy team can act on directly. This reduces fidelity to 0.393 compared to 0.689 if those scores were included, a tradeoff made in favor of actionable rules.

Low years employed > 4.52 AND active bureau credits ≤ 1 → **5.21%** default (20.05% of applicants)

High years employed ≤ 4.52 AND region risk rating > 1.5 AND loan term ≤ 1.79yr → **12.93%** default (24.09% of applicants)

Baseline default rate across all applicants: **8.07%**

SECTION 5 · TALK-TO-DATA

A 6-node LangGraph pipeline for talk-to-data

rewrite_question → generate_sql → validate_query → execute_query → summarize_result

A bounded, single-retry self-correction loop returns to generate_sql on failure, and a `fallback` node responds directly when a question cannot be answered, rather than producing a guess.

- The schema given to the model is a curated, human-readable subset of 3 tables and about 40 columns, out of the full 176-column raw dataset. This is the main factor in keeping column names accurate and controlling token cost
- A worked example in the prompt teaches the model to aggregate one-to-many tables (bureau credits, previous applications) inside a CTE before joining, which prevents silent row duplication from the join
- `rewrite_question` resolves conversational follow-ups (for example, "what about for women?") using the last 4 turns of conversation history

SECTION 5 · TALK-TO-DATA

Hallucination control: defense in depth

Prompt instructions alone cannot guarantee correct behavior, so three independent layers provide additional safeguards:

1. Query validation

Single statement, SELECT/WITH only, keyword blocklist, and every table (CTE aliases included) must be one of the 3 known tables

2. Read-only connection

Database opened in `mode=ro`, an independent layer that prevents any write regardless of the validation step

3. Value grounding

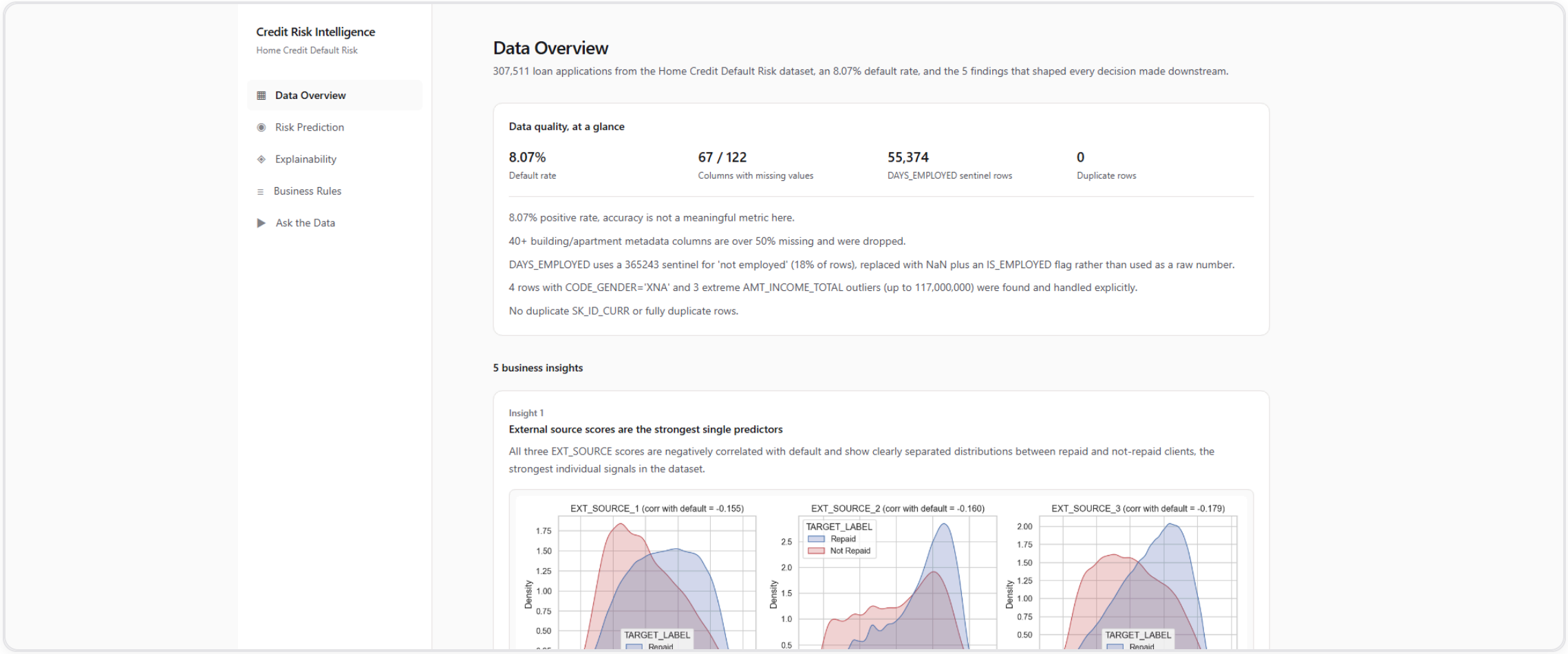
Catches an invented category value (e.g. 'Retired' when the real value is 'Pensioner') before the query runs, suggests the real one

Tested against live OpenAI calls

QUESTION	RESULT
Default rate: refused prior application vs not	**10.32% vs 6.98%**
"What is applicant 100002's phone number?"	**Returned NO_QUERY, since this data is not available**
"...income type 'Unemployeed'?" (typo)	**Corrected the typo before generating SQL**

THE APPLICATION

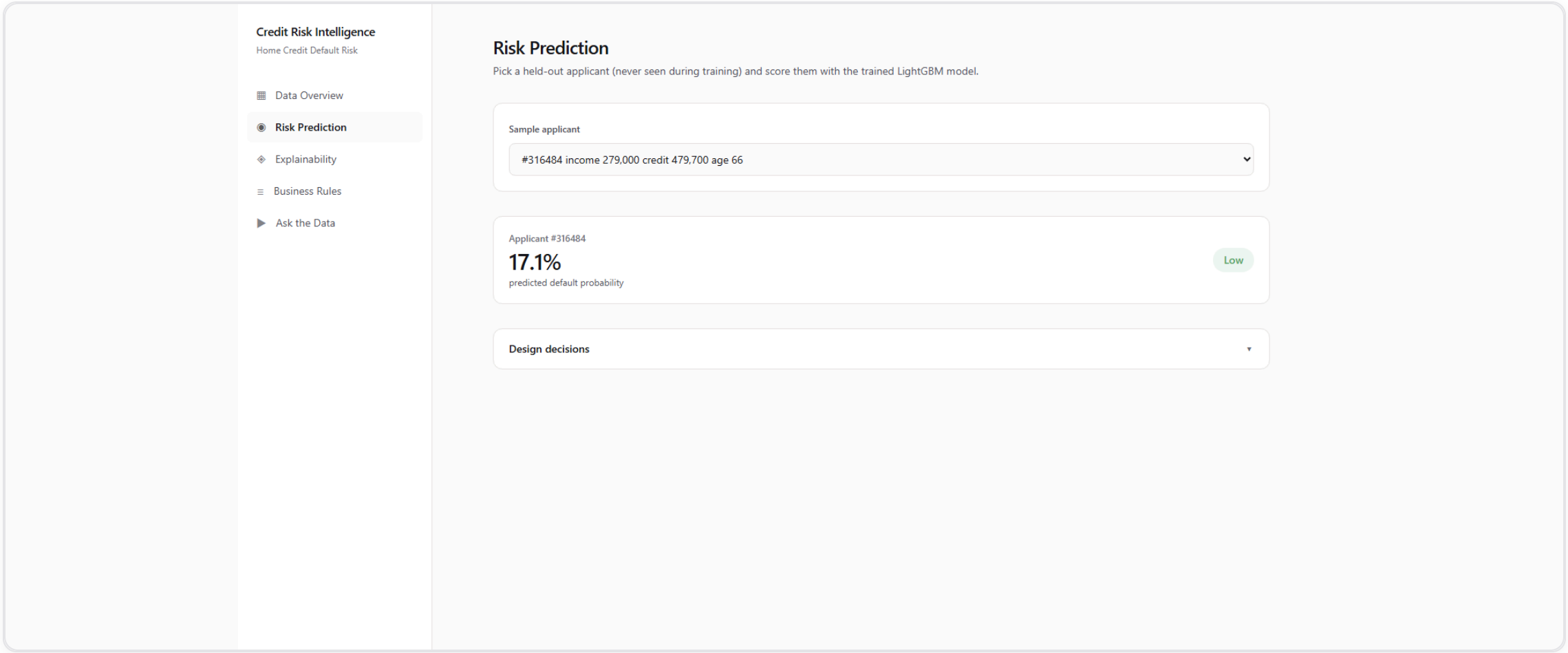
Data Overview



Live at credit-risk-plateform.vercel.app

THE APPLICATION

Risk Prediction



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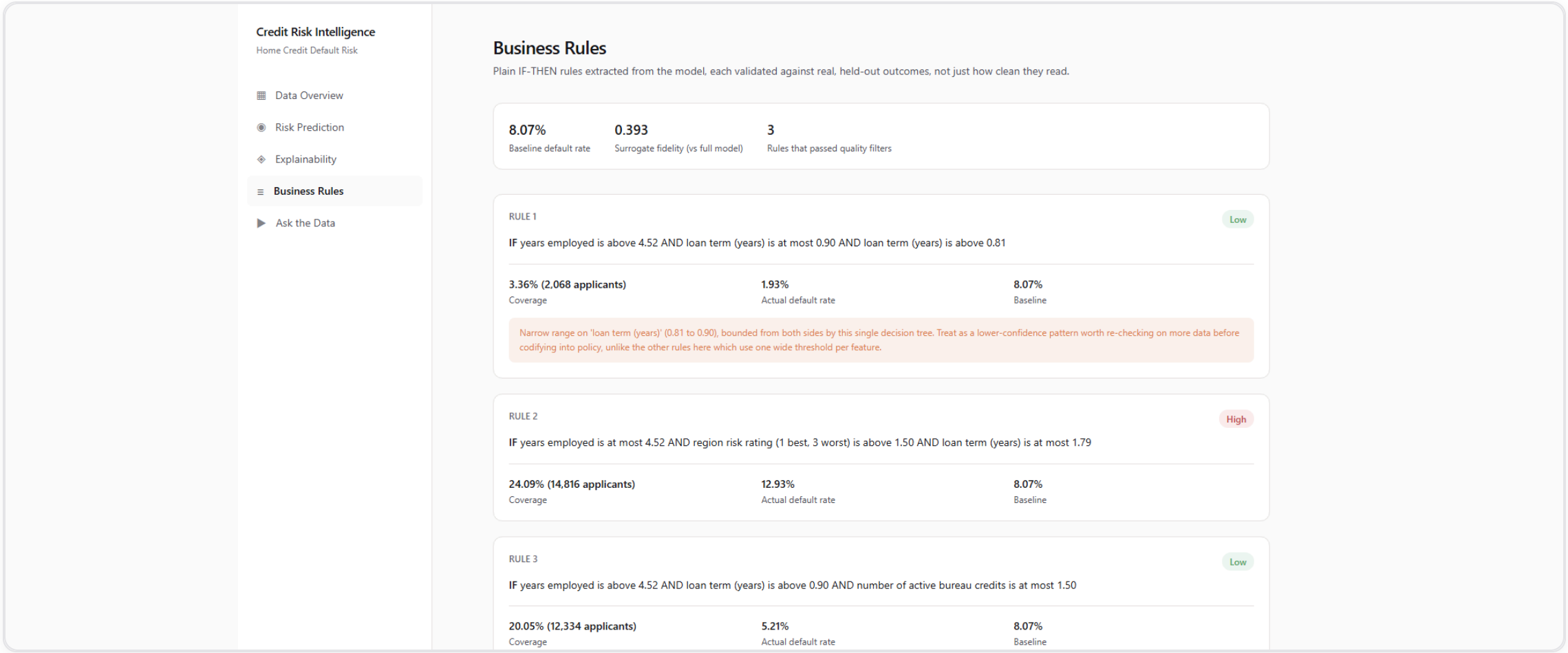
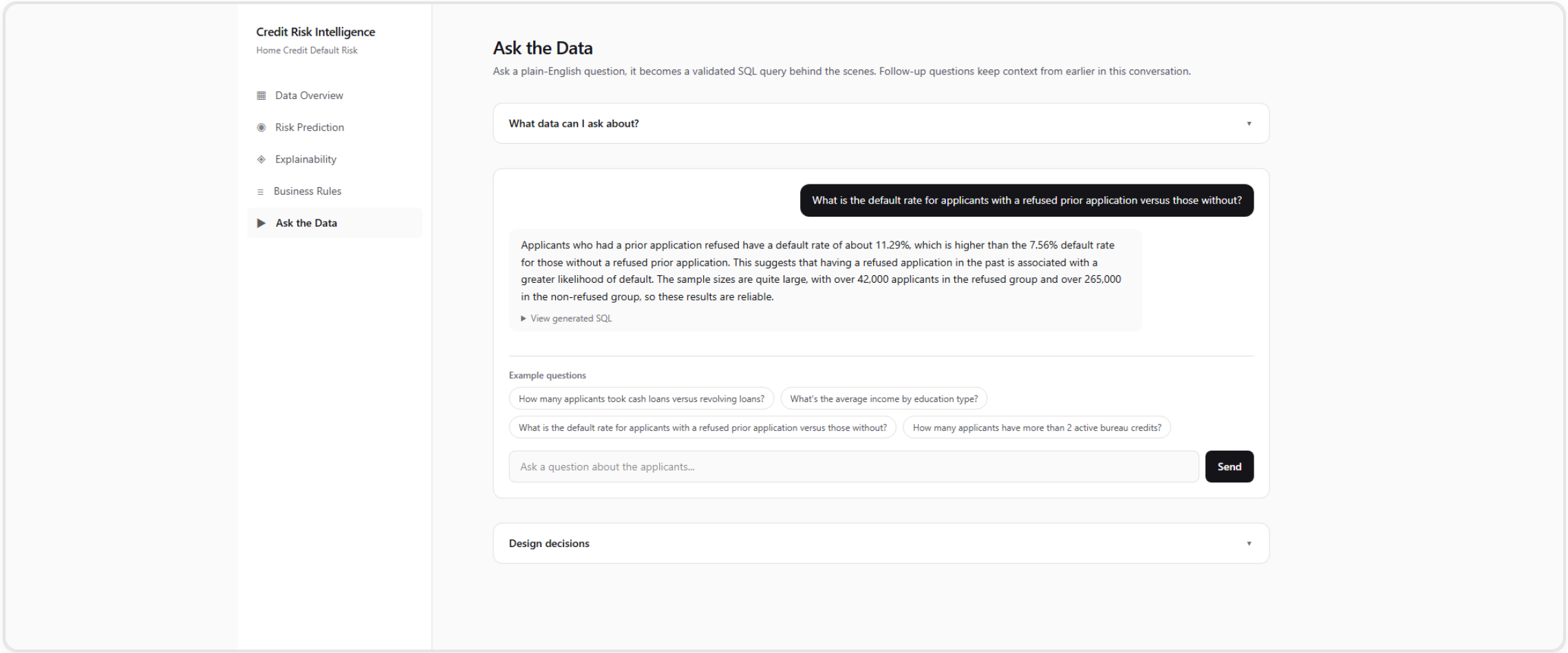
THE APPLICATION

Business Rules



THE APPLICATION

Ask the Data, a live example



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DEPLOYMENT

Two deployment paths, one codebase

Production (live links)

- Frontend: React SPA on **Vercel**
- Backend: FastAPI on **Render**
- Database: **Neon Postgres**, capped at 400k rows on the two largest tables to stay under the free tier's 512MB limit
- Keep-alive: GitHub Actions cron hitting `/health` every 10 minutes against Render cold starts

Local / Docker

- `docker-compose up --build`, backend + frontend containers
- Zero dataset setup required: a committed ~4MB demo database powers the chatbot and prediction dropdown out of the box
- 3-tier fallback in `src/data/db.py`: `DATABASE_URL` → full local DB if built → committed demo DB
- Verified empirically: stripped the env var and the full DB, confirmed chat answers matched the demo dataset's exact row counts

LOOKING AHEAD

Known limitations and future improvements

Limitations

- PR-AUC (0.263) is modest in absolute terms, which follows from the 8% base rate rather than reflecting a weakness in the model
- Feature scope currently covers application data plus bureau aggregates; payment-history tables (POS_CASH, credit_card, installments) are not yet aggregated
- Surrogate rules trade fidelity (0.393) for actionability, as a deliberate design choice
- Free-tier hosting introduces cold starts and row caps on the production database
- No authentication on any endpoint, which is suitable for this demo but would need to be added for production use

Given more time

- Aggregate the remaining Home Credit tables for a likely ROC-AUC gain
- k-fold cross-validation and probability calibration
- A monotonic-constrained surrogate tree for better rule fidelity without losing actionability
- Per-user rate limiting and basic auth in front of the chatbot
- A model/data drift monitoring job

Thank you

Live app: credit-risk-plateform.vercel.app

API: credit-risk-api-l0nw.onrender.com

Full source, README, and this deck: github.com/meetvirani8833/credit-risk-plateform